$\SHEAFHOM$ IS LEFT EXACT IN EACH VARIABLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate and re-verified against the pinned source (sm-twins-20261014.json span sanity: all OK).

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found hard occupation only from the 47 pre-existing lane-s11 S11-sites-mod-* cards (same pinned snapshot, delivered) and cross-chapter citation noise, 0 other-lane hits for sites-modules.tex (sm-occupied.json, 2026-10-14); the RESUME-pool dispatch unlocks the remaining face.

Source reference. Original lemma 03EN: sources/stacks-chow/tags.txt, source lines 4462–4462. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 010N: sources/stacks-chow/tags.txt, source lines 1333–1333. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{F}$, $\mathcal{G}$ be $\mathcal{O}$-modules.*

(1) *If $\mathcal{F}_2 \to \mathcal{F}_1 \to \mathcal{F} \to 0$ is an exact sequence of $\mathcal{O}$-modules, then*
$$0 \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}_1, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}_2, \mathcal{G})$$
is exact.

(2) *If $0 \to \mathcal{G} \to \mathcal{G}_1 \to \mathcal{G}_2$ is an exact sequence of $\mathcal{O}$-modules, then*
$$0 \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}) \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}_1) \to \mathcal{H}om_{\mathcal{O}}(\mathcal{F}, \mathcal{G}_2)$$
is exact.

Proof. Follows from Lemma 03EN and Homology, Lemma 010N. □

Date: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.